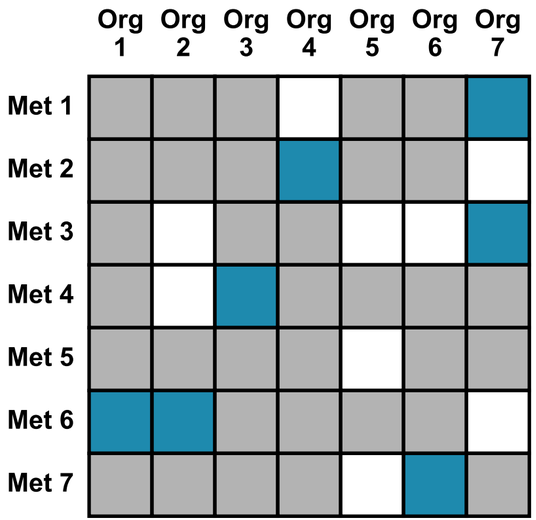
C

Input: assertion matrix

433 isolates × 311 metabolites;
no other features



depleted (positive) · neutral (measured negative) · unobserved (untested or secreted)

BPR training

sampled triples (i, j, k)

i isolate
j metabolite depleted by i
k neutral (50%) or unobserved (50%)

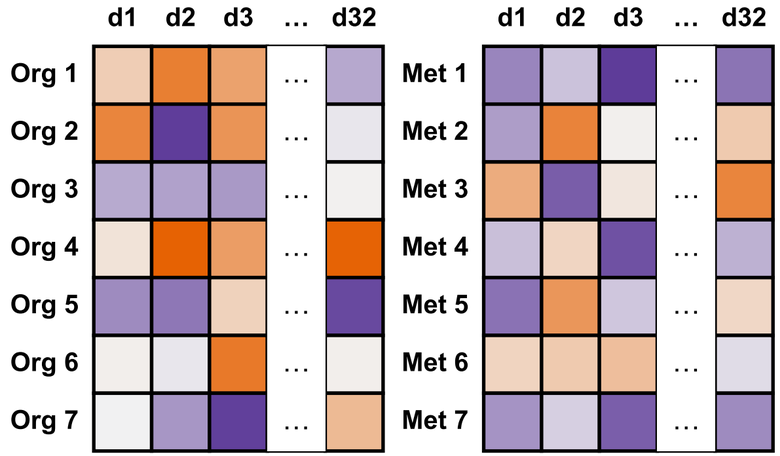
push $\mathbf{u}_i \cdot \mathbf{v}_j > \mathbf{u}_i \cdot \mathbf{v}_k$

Isolate embeddings

u (32 values each)

Metabolite embeddings

v (32 values each)



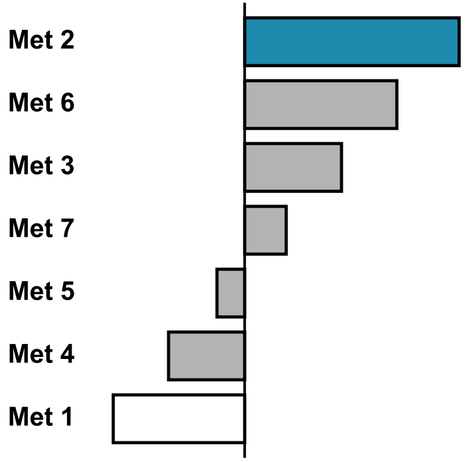
− + learned values (illustrative)

score

$\mathbf{u}_i \cdot \mathbf{v}_j$

Every pair scored

ranking for Org 4
(illustrative)



unobserved metabolites (grey) are ranked with measured ones